**1** Profit = \$400 − \$320 = \$80 or $\frac{400}{320} \times 100$

[M1]

25 **[A1]**

*[Total Marks: 2]*

**2 (a)** $\frac{24}{60} \times 100$ **[M1]**

40 **[A1]**

**(b)** $\frac{15}{100} \times 80 = 12$ OR $80 \times 0.85$ **[M1]**

68 **[A1]**

*[Total Marks: 4]*

**3 (a)** $\frac{15}{100} \times 320$ or $10\% = 32, 5\% = 16$ **[M1]**

48 **[A1]**

**(b)** $\frac{20}{100} \times 45 = 9$ or $\frac{80}{100} \times 45$ **[M1]**

$45 - 9$ (if first method used) **[M1]**

36 **[A1]**

*[Total Marks: 5]*

**4 (a)** Discount $= 0.15 \times 120 = 18$ OR $0.85 \times 120$ **[M1]**

102 **[A1]**

**(b)** Let cost price be $C$. $1.2C = 120$ **[M1]**

$C = \frac{120}{1.2}$ **[M1]**

100 **[A1]**

*[Total Marks: 5]*

**5 (a)** $4500 \times \left(1 + \frac{3.2}{100}\right)^5$ **[M2]**

OR $4500 \times 1.032^5$ **[M2]**

(M1 for $4500 \times 1.032^k$ where $k > 1$)

5267.57... **[A1]**

**(b)** $\frac{408 - 320}{320} \times 100$ or $\frac{88}{320} \times 100$ **[M1]**

27.5 **[A1]**

*[Total Marks: 5]*

**6** **(a)** $45\,000 \times 1.042$ OR $45\,000 + 45\,000 \times 0.042$ **[M1]**

$46\,890$ **[A1]**

**(b)** $100\% - 24\% = 76\%$ **[B1]**

$\frac{342}{0.76}$ OR $\frac{342}{76} \times 100$ **[M1]**

$450$ **[A1]**

*[Total Marks: 5]*

**7** **(a)** $4500 \times (1 + \frac{2.4}{100})^5$ or $4500 \times 1.024^5$ **[M2]**

If M0, M1 for $4500 \times 1.024$ or 4608 seen.

$5066.57$ or $5066.56\ldots$ **[A1]**

**(b)** Interest earned $= 5066.57 - 4500 = 566.57$ (accept unrounded) **[M1]**

$\frac{4500 \times r \times 5}{100} = 566.57$ OR $225r = 566.57$ **[M1]**

$2.52$ or $2.518\ldots$ **[A1]**

*[Total Marks: 6]*

**8** **(a)** $4500 \times \left(1 + \frac{2.4}{100}\right)^5$ or $4500 \times 1.024^5$ **[M2]**

[M1] for $4500 \times 1.024^n$ where $n > 1$

$5066.55$ **[A1]**

**(b)** $100\% - 18\% = 82\%$ or $0.82$ **[B1]**

$\frac{557.60}{0.82}$ or $x \times 0.82 = 557.60$ **[M1]**

$680$ **[A1]**

*[Total Marks: 6]*

**9** **(a)** $(1.25 \times 10^6) \times 1.045^{10}$ **[M1]**

$1941214.7...$ **[A1]**

$1.94 \times 10^6$ **[A1]**

**(b)** $299.25 = 86\%$ or $\frac{299.25}{1 - 0.14}$ **[M1]**

$348$ **[A1]**

**(c)** Let $V$ be the original value. $V \times 0.85 \times 0.90^2 = 17212.50$ **[M1]**

$V = \frac{17212.50}{0.85 \times 0.81} = 25000$ **[A1]**

*[Total Marks: 7]*

**10** **(a)** $1\,250\,000 \times (1 + \frac{x}{100})^3 = 1\,367\,631$ **[M1]**

$(1 + \frac{x}{100})^3 = 1.0941048$ **[M1]**

$1 + \frac{x}{100} = \sqrt[3]{1.0941048} = 1.0304\cdots \implies x = 3.04\cdots \approx 3$ **[A1]**

**(b)** Target population $= 2 \times 1\,250\,000 = 2\,500\,000$ **[B1]**

$1\,250\,000 \times (1.03)^n > 2\,500\,000 \implies 1.03^n > 2$ **[M1]**

Using logs: $n\log(1.03) > \log(2)$ OR trial and improvement yielding $n$ between 23 and 24 **[M1]**

$n > 23.449\ldots$, so it takes 24 years. Year $= 2015 + 24 = 2039$ **[A1]**

*[Total Marks: 7]*

**11** **(a)** $5000 \times \left(1 + \frac{2.5}{100}\right)^3$ **[M1]**

5384.45 **[A1]**

**(b)** $\left(1 + \frac{r}{100}\right)^5 = 1.30$ **[M1]**

$1 + \frac{r}{100} = \sqrt[5]{1.30}$ **[M1]**

5.39 (or $5.387\ldots$) **[A1]**

**(c)** $V \times (1 - 0.12)^3 = 17\,036.80$ **[M1]**

$V = \frac{17\,036.80}{0.88^3}$ **[M1]**

25 000 **[A1]**

*[Total Marks: 8]*

**12** **(a)** Expression for 3 years depreciation: e.g. $V \times (1 - 0.15) \times (1 - 0.08)^2$ or $V \times 0.85 \times 0.92^2$ **[M2]**

[M1] for $V \times 0.85$ or $V \times 0.92^2$ seen

$V \times 0.71944 = 17986$ or $V = \frac{17986}{0.85 \times 0.92^2}$ **[M1]**

25 000 **[A1]**

**(b)** $\frac{25000 - 17986}{25000} \times 100$ or $100\% - (0.85 \times 0.92^2 \times 100)\%$ **[M1]**

28.1 **[A1]**

**(c)** $P \times \left(1 + \frac{r}{100}\right)^{12} = 2P$ or $\left(1 + \frac{r}{100}\right)^{12} = 2$ **[M1]**

$1 + \frac{r}{100} = \sqrt[12]{2}$ or 1.059... **[M1]**

5.95 (correct to 3 significant figures) **[A1]**

*[Total Marks: 9]*